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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025
FIRST SEMESTER
Multi-Disciplinary Courses (MDC)-MG1MDCBIF100-BIOINFORMATICS FOR BEGINNERS
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 35

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest() and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A
Short Answer Questions
Answer any & out of 7 Questions
Each Question carries 2 marks

1. Interpret hierarchical organization of life. [U]/[CO1, CO2]
   - It means that living organisms are structured in levels, starting from the smallest (atoms → molecules → cells) and building up to larger, more complex levels (tissues → organs → organisms → ecosystems). Each level is made of the level below it and works together to form the complete living system.

2. Relate the purpose of global alignment in bioinformatics [A]/[CO6]
   - It is used to compare two biological sequences end-to-end to see how similar they are across their entire length. Its purpose is to identify overall similarity, evolutionary relationships, and conserved regions between two full sequences.

3. Which organization manages the Protein Data Bank (PDB)? [K]/[CO5, C06]
   - It is managed by the Worldwide Protein Data Bank (wwPDB), a global consortium that includes RCSB PDB (USA), PDBe (Europe), PDBj (Japan), and BMRB (USA).

4. Name the primary function of Clustal Omega. [K]/[C06]
   - The primary function of Clustal Omega is to perform multiple sequence alignment (MSA) of protein or nucleotide sequences.

5. Draw a simple representation of a DNA sequence with its complementary strand. [K]/[C06]
   - Any representation accurately showing the template and complementary strand

6. Cherish the important process involved in central Dogma of biology. [Ap]/[C01,CO2]
   - Transcription – DNA is copied into mRNA.
   - Translation – mRNA is decoded to produce proteins.

7. How has the Human Genome Project impacted medicine? (K)/[C04]

- It transformed medicine by enabling personalized and predictive healthcare, helped identify genes linked to diseases, improved diagnostic accuracy, guided targeted drug development, and allowed early detection and prevention based on a person's genetic risk.

[5x210]

Part B

Short Essay Type Questions

Answer any 3 out of 5 Questions

Each Question carries 5 marks.

8. Correlate the role of bioinformatics in drug discovery and development. [An]/[CO6]

- Introduction to bioinformatics in drug discovery – 1 mark
- Target identification & validation – 1 mark
- Virtual screening – 1 mark
- ADMET prediction – 1 mark
- Overall significance in faster, cost-effective drug development – 1 mark

9. Uphold the aim and scope of bioinformatics. [Ap]/[CO3]

- Aim of Bioinformatics (2 Marks)
- Scope of Bioinformatics (3 Marks)

10. Ilustrate the process of retrieving data from UniProt? [U]/[C05,C06]

- Accessing UniProt website – 1 mark
- Entering the protein name/ID in the search bar – 1 mark
- Selecting the correct entry from results – 1 mark
- Viewing protein details (sequence, function, structure) – 1 mark
- Downloading sequences/data (FASTA, features, etc.) – 1 mark

11. Define pairwise sequence alignment and explain its significance in bioinformatics. [U]/ [CO6]

- Definition of pairwise sequence alignment – 1 mark
- Types (global/local) or basic explanation – 1 mark
- Purpose of comparing two sequences – 1 mark
- Significance in identifying similarity/evolution/functional relationships – 1 mark
- Applications in bioinformatics (e.g., gene prediction, homology, drug design) – 1 mark

12. Analyze the significance of structural databases in understanding protein function and interactions . [U]/[CO5, CO6]

- Definition of structural databases – 1 mark
- Role in understanding 3D protein structure – 1 mark
- Importance in predicting protein function – 1 mark
- Use in studying protein–protein interactions – 1 mark

- Applications in drug design/biomedical research – 1 mark

[3x5 = 15]

Part C

Long Essay Type Questions
Answer any 1 out of 3 Questions
Each Question carries 10 marks

13. Define molecular visualization and explain its significance in bioinformatics. [U]/ [CO6]

- Definition of molecular visualization – 2 marks
- Explanation of basic purpose (visualizing 3D structures) – 1 mark
- Role in understanding protein structure and function – 2 marks
- Use in analyzing molecular interactions (ligand binding, active sites) – 2 marks
- Applications in drug discovery and structural biology – 2 marks
- Conclusion/overall significance – 1 mark

14. Explore biomolecules and its types. [I]/ [CO1, CO2]

- Definition of biomolecules – 2 marks
- Classification of biomolecules – 2 marks
- Carbohydrates (types & functions) – 2 marks
- Proteins (types & functions) – 2 marks
- Lipids & Nucleic acids (overview) – 2 marks

15. Embrace the basic difference in computational biology and bioinformatics.[I]/ [CO3]

- Definition of Computational Biology – 1 mark
- Definition of Bioinformatics – 1 mark
- Key Differences in Focus/Approach – 4 mark
- Examples of Computational Biology methods – 2 marks
- Examples of Bioinformatics analysis– 2 marks

[1x10 = 10]